BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: KT-X-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code B

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. The resistance of wire will decrease by increasing.....

(A) temperature
(B) length
(C) diameter
(D) both A and B

Qii. Which derived unit is equivalent to tesla T?

(A) Nm/A
(B) Am/N
(C) NA/m
(D) N/Am

Qiii. The Boolean equation for OR gat is .....

(A) X = AB
(B) X = A+B
(C) X = AB (whole overlined)
(D) X = A+B (whole overlined)

Qiv. Cathode rays are beam of ....

(A) protons
(B) electrons
(C) ions
(D) neutrons

Qv. Electric or electromagnetic representations of data or information is termed as .....

(A) computer
(B) signal
(C) transmission
(D) software

Qvi. ^{39}Ar is an isotope with a half life of 269 years, it will reduce to half in .....

(A) 134.5 minutes
(B) 269 minutes
(C) 134.5 years
(D) 269 years

Qvii. A certain pendulum has an iron bob. When the iron bob is replaced with lead bob of the same size, its time period will.....

(A) stay the same
(B) decrease
(C) increase
(D) be zero

Qviii. The bending of waves around the edges of the obstacle is .....

(A) reflection
(B) refraction
(C) diffraction
(D) damping

Qix. Minimum echo distance is reduced in.....

(A) summer
(B) winter
(C) spring
(D) space

Qx. Which one of the following material will refract light more.

(A) water
(B) glass
(C) air
(D) diamond

Qxi. When combing our hair, we shift electrons from our hair onto the comb, the charge on our hairs is

(A) positive
(B) negative
(C) zero
(D) infinite

Qxii. The unit of electric potential is.....

(A) Farad
(B) Coulomb
(C) Volt
(D) NC^{-1}

Section "B"

Time Allowed: 2 Hours 40 minutes | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i Suppose you stand on a swing instead of sitting on it. Will your frequency of oscillation increase or decrease?

ii Why sound produced by a simple pendulum is not heared?

iii Define Snell's law with mathematical form.

iv An external rear view car mirror is convex with a radius of curvature of 16m determine the focal length of the mirror.

v How does shuffling feet across a carpet causes hair to stand on our body?

vi Calculate the resistance of a 40W automobile head light designed for 12V.

vii Why does a compass needle points north?

viii What conditions produce a high (1) output for an AND gate and NOR gate?

ix Can internet be used for shopping? Give 3 examples.

x The atomic number of one particular isotope is equal to its mass number. Which isotope is it?

xi Define capacitor and its unit.

Section "C"

Maximum Marks: 21

Note: Note: Attempt any THREE of the following questions. All questions carry equal marks:

QQ.3.

a [4] Describe the terms loudness and pitch by giving an example.

b [3] Show that if one sound is twice as intense as another. About what factor is the sound level (in dB) higher?

QQ.4.

a [4] What is meant by reflection of light? State and explain laws of reflection with diagrams.

b [3] A 30V battery is connected to a 10Ω resistor. What is the current in the circuit?

QQ.5.

a [4] Explain the concept of electric energy and electric power. What is the commercial unit for the consumption of electric energy?

b [3] A 1.5m long wire carries a current of 5A, at right angle to a uniform magnetic field of 0.04T. Determine the force exerted on the wire.

QQ.6.

a [4] What is radioactivity? Give the nature, ionizing and penetrating abilities of the three types of radioactive emissions.

b [3] Define logic gates. Describe the operation of NAND gate by drawing their symbol and truth table.